A Five-Point Counterexample to Concavity of Determinantal Entropy

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Abstract

Lyons asked whether the Shannon entropy of a finite determinantal probability measure is concave in its positive-contraction kernel. We give a counterexample on five labelled points. At a real symmetric kernel, every event probability is even along a purely imaginary Hermitian direction. The Fisher-information term in the entropy Hessian therefore vanishes, and the remaining curvature term can be positive. Four short integer vectors specify an example. For explicit strict positive contractions Q_+, Q_- with midpoint K, an exact rational calculation proves $H(Q_+) = H(Q_-) > H(K)$.

1 The question and the answer

Let $E = [5]$. If K is a Hermitian matrix with $0 \preceq K \preceq I$, the determinantal probability measure $\mathbb{P}^K$ on subsets of E is defined by

$$\mathbb{P}^K(T \subseteq X) = \det K[T].$$

Writing $p_S(K) = \mathbb{P}^K(X = S)$, its entropy is

$$H(K) = -\sum_{S \subseteq E} p_S(K) \log p_S(K).$$

The conjecture that H is concave in K was posed by Lyons [1] and restated as Conjecture 2.6 in his ICM survey [2]. The formulation there permits real or complex Hilbert spaces.

Here is the counterexample. Put

$$\mathbf{1} = (1,1,1,1,1)^T, \qquad x = (-2,-1,0,1,2)^T,$$
$$q_2 = (2,-1,-2,-1,2)^T, \quad q_3 = (-1,2,0,-2,1)^T,$$

and write $u \wedge v = uv^T - vu^T$. Define

$$P = \frac{1}{5}\mathbf{1}\mathbf{1}^T + \frac{1}{10}xx^T, \qquad B = \mathbf{1} \wedge x - \frac{1}{2} q_2 \wedge q_3, \qquad A = iB, \tag{1}$$

and

$$\varepsilon = 10^{-6}, \qquad K = \varepsilon I + (1 - 2\varepsilon)P, \qquad t = 10^{-9}, \qquad Q_\pm = K \pm tA. \tag{2}$$

Theorem 1. *The matrices in (1)–(2) satisfy $0 < Q_\pm < I$, and their complete event distributions obey*

$$p_S(Q_+) = p_S(Q_-) \quad (S \subseteq E), \qquad 2.81518 \times 10^{-19} < H(Q_+) - H(K) < 2.81520 \times 10^{-19}.$$

In particular,

$$H(K) < \frac{H(Q_-) + H(Q_+)}{2}.$$

Thus Conjecture 2.6 is false as stated.

The same example has a positive entropy Hessian at its midpoint:

$$\frac{1}{2} < D^2 H_K[A, A] < \frac{3}{5}, \qquad D^2 H_K[A, A] = 0.56612746277851002589\ldots.$$

2 Why a purely imaginary direction is useful

The complete-event probabilities admit the signed determinant formula

$$p_S(K) = (-1)^{|E \setminus S|} \det\bigl(K - I_{E \setminus S}\bigr), \tag{3}$$

where I_T is the diagonal coordinate projection onto T. This follows directly from Möbius inversion of the inclusion probabilities.

Let $\Phi(K) = (p_S(K))_S$ be the map from kernels to the probability simplex. At an interior kernel, differentiating entropy twice gives

$$D^2 H_K[A, A] = -\sum_S \frac{(Dp_S(K)[A])^2}{p_S(K)} - \sum_S D^2 p_S(K)[A, A] \log p_S(K). \tag{4}$$

The first sum is the pullback of the Fisher metric and is always non-positive. The second sum records the curvature of the probability map in the direction of the entropy gradient.

The construction makes the first sum disappear. If K is real symmetric and B is real skew-symmetric, then

$$\det(K - I_{E \setminus S} + itB) = \det(K - I_{E \setminus S} - itB),$$

because transposition changes t to $-t$. Thus every event probability is an even function of t, and

$$Dp_S(K)[iB] = 0 \qquad (S \subseteq E). \tag{5}$$

If

$$b_S(K, B) = \frac{d^2}{dt^2} p_S(K + itB)\bigg|_{t=0},$$

then (4) reduces to

$$D^2 H_K[iB, iB] = -\sum_S b_S(K, B) \log p_S(K). \tag{6}$$

In information-geometric language, iB lies in the differential kernel of Φ; the sign is decided entirely by the vertical second-order bending of Φ.

3 How the four vectors were chosen

The vectors $\mathbf{1}, x, q_2, q_3$ are pairwise orthogonal across the two displayed pairs: $\mathbf{1} \perp x$, and $q_2, q_3 \perp \mathbf{1}, x$. Since $\|\mathbf{1}\|^2 = 5$ and $\|x\|^2 = 10$, the matrix P is the orthogonal projection onto $\langle \mathbf{1}, x \rangle$. The two summands of B rotate, respectively, inside $\mathrm{ran}(P)$ and inside $\ker(P)$. Hence

$$P^2 = P = P^T, \qquad B^T = -B, \qquad PB = BP. \tag{7}$$

It follows at once that K has eigenvalue $1 - \varepsilon$ on $\mathrm{ran}(P)$ and eigenvalue ε on $\ker(P)$. Thus $0 < K < I$, while $A = iB$ is Hermitian.

Both summands of B commute with P. This commutation, rather than the coefficient $-\frac{1}{2}$, is what removes the logarithmic divergence in the boundary calculation below. The limiting curvature is an indefinite quadratic form on the two-dimensional space spanned by these rotations. The coefficient $-\frac{1}{2}$ is a simple rational choice inside its positive cone.

4 The positive boundary limit

For $0 < \delta < 1/2$, set $K_\delta = \delta I + (1-2\delta)P$. The coordinates of x are distinct, so every two-row minor of the matrix with columns $\mathbf{1}, x$ is nonzero. Consequently P represents the uniform matroid $U_{2,5}$: every two-coordinate principal minor is positive. For each subset S there is a positive rational number a_S such that

$$p_S(K_\delta) = a_S \delta^{w_S}(1 + O(\delta)), \qquad w_S = \big||S| - 2\big|. \tag{8}$$

More explicitly,

$$a_S = \det P[S] \quad (|S| \le 2), \qquad a_S = \det(I - P)[E \setminus S] \quad (|S| \ge 2).$$

These formulas follow either from the spectral Bernoulli mixture for K_δ, or directly from principal-minor expansion and Jacobi's complementary-minor identity.

Let $b_S = b_S(P, B)$. Direct second-order determinant expansion gives the two exact cancellations

$$\sum_S b_S = 0, \qquad \sum_S w_S b_S = 0. \tag{9}$$

The first is forced by $\sum_S p_S = 1$. The second follows here from $PB = BP$. Equations (6)–(9), together with $\delta \log \delta \to 0$, now give

$$\begin{aligned} \lim_{\delta \downarrow 0} D^2 H_{K_\delta}[iB, iB] &= -\sum_S b_S \log a_S \\ &= \frac{1}{5} \log \frac{2^{260} 3^{143} 5^{350}}{7^{87} 13^{284}} = 0.57665861803438136898\ldots. \end{aligned} \tag{10}$$

This is an exact positive number: the numerator inside the logarithm is a 392-digit integer, whereas the denominator is a 390-digit integer.

The role of the coefficient in (1) can now be seen directly. Let $B_1 = \mathbf{1} \wedge x$ and $B_2 = q_2 \wedge q_3$. Each direction separately has a finite, negative boundary limit:

$$\lim_{\delta \downarrow 0} D^2 H_{K_\delta}[iB_1, iB_1] = -24.38211062\ldots, \qquad \lim_{\delta \downarrow 0} D^2 H_{K_\delta}[iB_2, iB_2] = -72.60746516\ldots.$$

Their mixed term makes the quadratic form indefinite, and the rational direction $B_1 - \frac{1}{2}B_2$ gives the positive value in (10). It was chosen for this simple exact form, not to maximize the normalized curvature.

For transparency, Table 1 groups the 32 determinant terms by their value of a_S. Its second column is $B_a = \sum_{S: a_S = a} b_S$; hence $-\sum_a B_a \log a$ reproduces (10). The ungrouped table is supplied with the checker.

Table 1: Grouped data for the boundary logarithmic sum.

a	B_a	a	B_a
1/50	72/5	2/5	−2
2/25	−204/5	12/25	128/5
3/25	41/5	13/25	284/5
9/50	−606/5	3/5	120
1/5	20	7/10	−32
7/25	247/5	4/5	−36
3/10	60	1	−100
8/25	−112/5		

5 An exact interior certificate

The positive limit proves that all sufficiently small δ have positive curvature in this direction, but the scale is worth recording. The same exact interval calculation gives

$$D^2 H_{K_{10^{-4}}}[iB, iB] < 0, \qquad D^2 H_{K_{10^{-5}}}[iB, iB] > 0.$$

Their numerical values are $-0.0710698\ldots$ and $0.4916113\ldots$, respectively; the zero lies numerically near 8.7430×10^{-5}. This numerical location is not used in the proof. We fix $\varepsilon = 10^{-6}$ and verify both the positive Hessian and a finite midpoint violation using rational arithmetic throughout.

For a positive rational $r \in [1, 2]$, set $y = (r-1)/(r+1)$. The positive series for $2\,\mathrm{atanh}(y)$ gives, for every positive integer m,

$$2\sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1} \le \log r \le 2\sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1} + \frac{2y^{2m+1}}{(2m+1)(1-y^2)}. \tag{11}$$

Every positive rational probability is written as $2^k r$ with $1 \le r < 2$. Taking $m = 24$ in (11) gives directed rational intervals for all logarithms in (6).

The event probabilities and second derivatives are also rational. They are computed by expanding each five-by-five determinant over its 120 permutations and extracting the coefficients of t^0 and t^2. Summation checks

$$p_S(K) > 0 \quad \text{for all } S, \qquad \sum_S p_S(K) = 1, \qquad \sum_S b_S(K, B) = 0.$$

Applying the lower logarithm endpoint to positive coefficients and the upper endpoint to negative coefficients, with the choices reversed for the upper bound, proves

$$0.566127462778510025889 3 < D^2 H_K[iB, iB] < 0.5661274627785100258923. \tag{12}$$

The endpoints in the computation are exact fractions; the decimals in (12) are shortened outward displays. In particular, $\frac{1}{2} < D^2 H_K[iB, iB] < \frac{3}{5}$.

For the finite chord, direct calculation gives $\|B\|_F^2 = 170$. Since $\lambda_{\min}(K) = \lambda_{\min}(I - K) = 10^{-6}$,

$$t\|B\|_{\mathrm{op}} \le t\|B\|_F < 14 \times 10^{-9} < 10^{-6}.$$

Thus $Q_\pm$ are strict positive contractions. The evenness in (5) holds for the full determinant polynomial, so $p_S(Q_+) = p_S(Q_-)$ exactly. Applying (11) directly to the center and endpoint distributions gives the directed rational enclosure

$$2.81518 \times 10^{-19} < H(Q_+) - H(K) < 2.81520 \times 10^{-19}.$$

This proves Theorem 1 in its midpoint form.

The accompanying file `verify_exact.py` is self-contained and uses only the Python standard library. Starting from the four vectors, it rebuilds all 32 boundary terms, the prime-factor expression in (10), the interior Hessian bounds, both endpoint distributions, and the finite entropy gap. It also gives an exact rational certificate for the real symmetric Hessian block described below. Its SHA-256 digest is

`FBEA3920226D096A9490DE89EFFD6A8BA39C5AC68EDCA9DE251E445E5520148C.`

6 Scope

The midpoint K is real symmetric, but the direction $A = iB$ is purely imaginary. Complex conjugation fixes real symmetric directions and negates purely imaginary Hermitian directions. Since $H(\overline{L}) = H(L)$, the two subspaces are orthogonal for the Hessian at a real kernel. The counterexample lies entirely in the imaginary block.

At the particular midpoint K above, the Hessian is negative definite on the entire real symmetric block. The checker proves this by an invertible rational change of basis after which the negative Hessian is strictly diagonally dominant. Thus the real restriction was tested at this kernel, although this one local calculation does not settle it in general.

The result therefore disproves Conjecture 2.6 with the real-or-complex quantifier used in the cited sources. It does not decide the separate question obtained by requiring all kernels and directions to be real symmetric, and it does not establish that five is the smallest possible dimension.

We make no absolute priority claim. To the best of our knowledge, the cited conjecture had no previously published proof or counterexample; private communications, seminars, unpublished notes, and unsuccessful attempts are outside the scope of our search.

References

[1] Russell Lyons. Determinantal probability measures. *Publications Mathématiques de l'IHÉS*, 98:167–212, 2003.

[2] Russell Lyons. Determinantal probability: Basic properties and conjectures. In *Proceedings of the International Congress of Mathematicians 2014*, volume IV, pages 137–161, 2014.